Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Title:	**DESIGN CALCULATION** **3710 TAILINGS PASTE THICKENING** **DESIGN OF STAIR FOUNDATION**
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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The purpose of this document is to analysis and design of **TAILINGS PASTE THICKENING** stair foundation. In this document design & analysis of foundation have been covered. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week

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3. SCOPE

The analysis has been carried out in Staad pro & designed by Excel Sheet. For member strength calculation force values are directly taken from STAAD Pro & designed in excel sheet. Load data for foundation design taken from document no. **MD-216-3710-EG-ST-CAL-2330.**

4. DESIGN BASIS

4.1 Material

A. Concrete Strength of 28 days

Foundation and Structural Members: 35 MPa

Blinding Concrete below foundation bases: 15 MPa

B. Reinforcement Grade

Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)

Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

C. Grouting

Grout shall be 50 mm cementitious of Strength 40 MPa.

4.2 Loading

A. Primary Load

LOAD 1 = DEAD LOAD (DL)

LOAD 2 = LIVE LOAD (LL)

LOAD 3 = OPERATING FLUID LOAD (FL)

LOAD 4 = EQP DL (EQDL)

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LOAD 5 = EQP LL (EQLL)

LOAD 6 = WIND LOAD (+X) WL(+X)

LOAD 7 = WIND LOAD (-X) WL(-X)

LOAD 8 = WIND LOAD (+Z) WL(+Z)

LOAD 9 = WIND LOAD (-Z) WL(-Z)

LOAD 10 = SEISMIC LOAD SL(+X)

LOAD 11 = SEISMIC LOAD SL(-X)

LOAD 12 = SEISMIC LOAD SL(+Z)

LOAD 13 = SEISMIC LOAD SL(-Z)

LOAD 14 = TEMPERATURE LOAD (TL)

LOAD 15 = PIPE LATERAL LOAD (PL)

4.3 Load combination

A. The following load combination are considered for member strength calculation.

101 1.4(DL+EQDL+FL)

102 1.2(DL+EQDL+FL)+1.6(LL+EQLL)

103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)

104 1.4(DL+EQDL+FL+PL)

105 1.2(DL+EQDL+FL+PL)+1.6(LL+EQLL)

106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)

107 1.2(DL+EQDL)+0.8WL(+X)

108 1.2(DL+EQDL)+0.8WL(-X)

109 1.2(DL+EQDL)+0.8WL(+Z)

110 1.2(DL+EQDL)+0.8WL(-Z)

111 1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)

112 1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)

113 1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQLL)

114 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)

115 1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5FL

116 1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+0.5FL

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117 1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+0.5FL

118 1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+0.5FL

119 0.9(DL+EQDL)+1.6WL(+X)

120 0.9(DL+EQDL)+1.6WL(-X)

121 0.9(DL+EQDL)+1.6WL(+Z)

122 0.9(DL+EQDL)+1.6WL(-Z)

123 0.9(DL+EQDL)+SL(+X)

124 0.9(DL+EQDL)+SL(-X)

125 0.9(DL+EQDL)+SL(+Z)

126 0.9(DL+EQDL)+SL(-Z)

B) The following load combination are considered for base pressure, sliding & overturning check.

201 DL+EQDL+FL

202 DL+EQDL+FL+LL+EQLL+TL

203 DL+EQDL+FL+0.75(LL+EQLL)+0.75TL

204 DL+EQDL+FL+PL

205 DL+EQDL+FL+PL+LL+EQLL+TL

206 DL+EQDL+FL+PL+0.75(LL+EQLL)+0.75TL

207 DL+EQDL+FL+WL(+X)

208 DL+EQDL+FL+WL(-X)

209 DL+EQDL+FL+WL(+Z)

210 DL+EQDL+FL+WL(-Z)

211 DL+EQDL+FL+0.7SL(+X)

212 DL+EQDL+FL+0.7SL(-X)

213 DL+EQDL+FL+0.7SL(+Z)

214 DL+EQDL+FL+0.7SL(-Z)

215 DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQLL)

216 DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQLL)

217 DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQLL)

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218 DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQLL)

219 DL+EQDL+FL+0.525SL(+X)+0.75(LL+EQLL)

220 DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQLL)

221 DL+EQDL+FL+0.525SL(+Z)+0.75(LL+EQLL)

222 DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQLL)

223 0.6(DL+EQDL)+WL(+X)

224 0.6(DL+EQDL)+WL(-X)

225 0.6(DL+EQDL)+WL(+Z)

226 0.6(DL+EQDL)+WL(-Z)

227 0.6(DL+EQDL)+0.7SL(+X)

228 0.6(DL+EQDL)+0.7SL(-X)

229 0.6(DL+EQDL)+0.7SL(+Z)

230 0.6(DL+EQDL)+0.7SL(-Z)

4.4 Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm except where welded wire fabric is used for ties and spirals the minimum rebar size shall be 10 mm.

4.5 Minimum concrete cover

	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation & Columns	50

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4.6 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 450 kN/m2 at 1.5m depth below EGL (RL 912.000) refer Table 5 - BH-14 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.7 Vertical Modulus of Subgrade Reaction

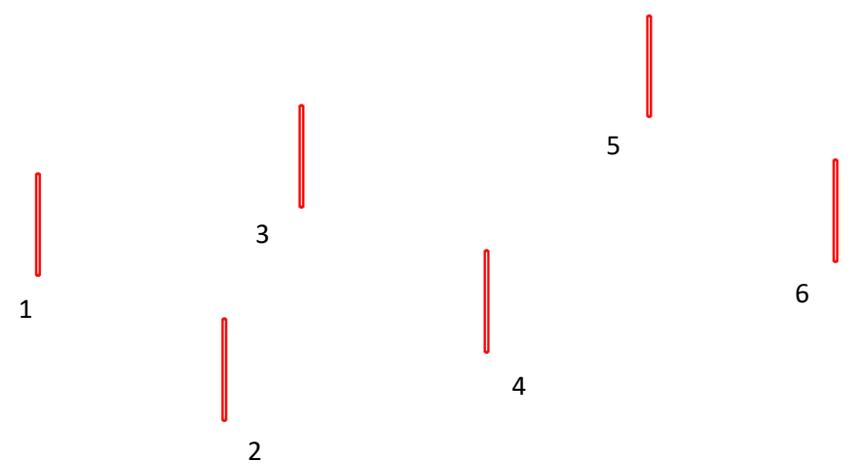
The Vertical Modulus of Subgrade Reaction has been considered as 54000 kN/m^2/m at 1.5m depth below EGL (RL 912.000) refer Table 5 - BH-14 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

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5.1 DESIGN OF FOUNDATION MKD. F1

(STAAD Node -1 2 3 4 5 6)

Size of Columns		Along-X	Along-Z
Node	1	0.55m	0.55m
Node	2	0.55m	0.55m
Node	3	0.55m	0.55m
Node	4	0.55m	0.55m
Node	5	0.55m	0.55m
Node	6	0.55m	0.55m
Size of fdn. raft:		12.30m	7.80m
Thickness of raft:		0.75 m	
Original Gr. Level:		0.00 m	
Finished Gr. Level:		0.00 m	
U/S of B/Pl Above GL		1.30 m	
U/S level of fdn:		-1.50 m	
Unit weight of conc:		25 kN/m^3	
Unit weight of soil:		18 kN/m^3	



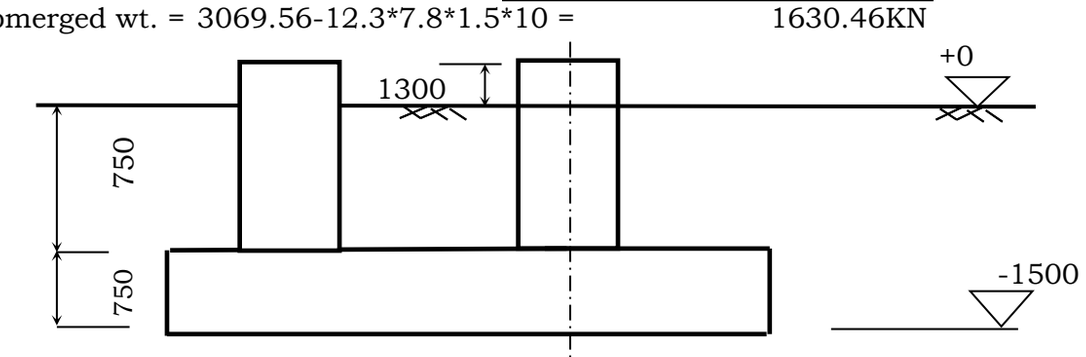
Calculation of Weight of Foundation :

Weight of raft = (12.3*7.8*0.75*25) = 1798.88KN

Weight of soil = ((12.3*7.8)-6*((0.55*0.55))*0.75*18 = 1270.69KN

Total = 3069.56KN

Submerged wt. = 3069.56-12.3*7.8*1.5*10 = 1630.46KN

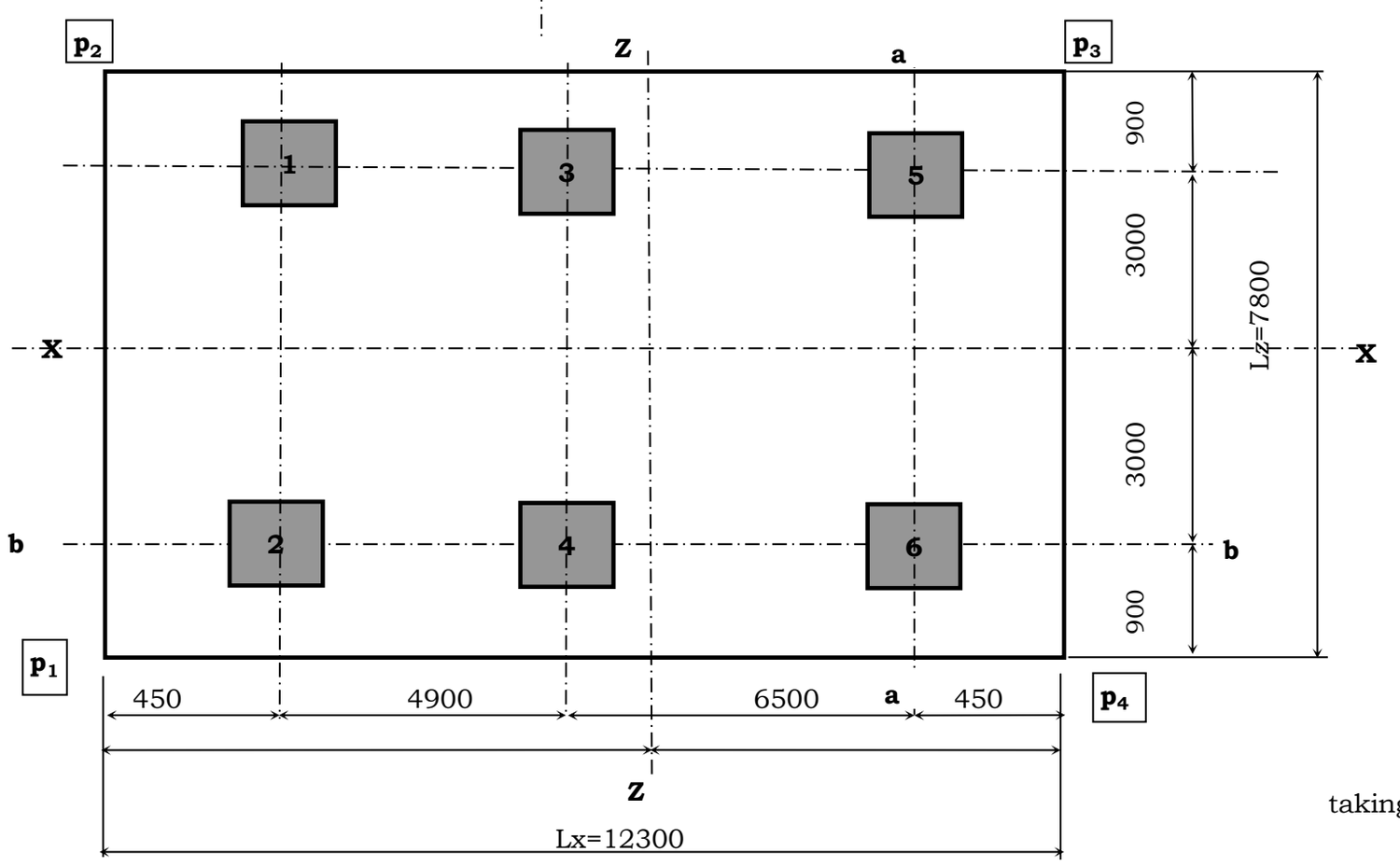


Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	16mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	16mm
Mininum percentage of reinforcement at bottom=	0.18%
Mininum percentage of reinforcement at top=	0.09%

Effective depth of slab along X direction = 684mm = 0.684m

Effective depth of slab along Z direction = 692mm = 0.692m

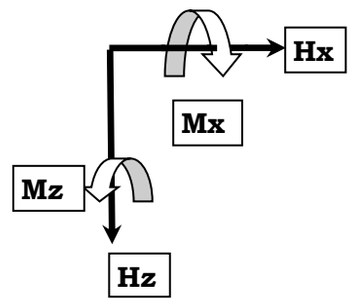


H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction



taking moment about a-a and b-b for eccentricity cal.

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Check for overturning and sliding of footing F1 :

Node	1					2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL+EQDL+FL	93.5	0	0	0	0	157.5	0	0	0	0
202 DL+EQDL+FL+LL+EQLL+TL	369.5	0	0	9	18.5	393.5	0	0	-3	-6.2
203 DL+EQDL+FL+0.75(LL+EQLL)+	300.5	0	0	6.8	13.8	334.5	0	0	-2.3	-4.6
204 DL+EQDL+FL+PL	25.5	0	0	0	0	206.5	0	0	-24	-49.2
205 DL+EQDL+FL+PL+LL+EQLL+T	301.5	0	0	9	18.5	442.5	0	0	-27	-55.4
206 DL+EQDL+FL+PL+0.75(LL+EQ	232.5	0	0	6.8	13.8	383.5	0	0	-26.3	-53.8
207 DL+EQDL+FL+WL(+X)	36.5	0	0	0	0	119.5	0	0	2	4.1
208 DL+EQDL+FL+WL(-X)	150.5	0	0	0	0	195.5	0	0	-2	-4.1
209 DL+EQDL+FL+WL(+Z)	-75.5	0	0	0	0	342.5	0	0	-64	-131.2
210 DL+EQDL+FL+WL(-Z)	262.5	0	0	0	0	-27.5	0	0	64	131.2
211 DL+EQDL+FL+0.7SL(+X)	32.6	0	0	0	0	111.3	0	0	1.4	2.9
212 DL+EQDL+FL+0.7SL(-X)	154.4	0	0	0	0	203.7	0	0	-1.4	-2.9
213 DL+EQDL+FL+0.7SL(+Z)	-61.9	0	0	1.4	2.9	330.4	0	0	-44.1	-90.4
214 DL+EQDL+FL+0.7SL(-Z)	248.9	0	0	-1.4	-2.9	-15.4	0	0	44.1	90.4
215 DL+EQDL+FL+0.75WL(+X)+0.7	257.8	0	0	5.3	10.8	306	0	0	-0.8	-1.5
216 DL+EQDL+FL+0.75WL(-X)+0.7	343.3	0	0	5.3	10.8	363	0	0	-3.8	-7.7
217 DL+EQDL+FL+0.75WL(+Z)+0.7	173.8	0	0	5.3	10.8	473.3	0	0	-50.3	-103
218 DL+EQDL+FL+0.75WL(-Z)+0.7	427.3	0	0	5.3	10.8	195.8	0	0	45.8	93.8
219 DL+EQDL+FL+0.525SL(+X)+0.	254.8	0	0	5.3	10.8	299.9	0	0	-1.2	-2.5
220 DL+EQDL+FL+0.525SL(-X)+0.7	346.2	0	0	5.3	10.8	369.2	0	0	-3.3	-6.8
221 DL+EQDL+FL+0.525SL(+Z)+0.	184	0	0	6.3	12.9	464.2	0	0	-35.3	-72.4
222 DL+EQDL+FL+0.525SL(-Z)+0.7	417.1	0	0	4.2	8.6	204.8	0	0	30.8	63.2
223 0.6(DL+EQDL)+WL(+X)	2.1	0	0	1.2	2.5	43.9	0	0	2	4.1
224 0.6(DL+EQDL)+WL(-X)	116.1	0	0	1.2	2.5	119.9	0	0	-2	-4.1
225 0.6(DL+EQDL)+WL(+Z)	-109.9	0	0	1.2	2.5	266.9	0	0	-64	-131.2
226 0.6(DL+EQDL)+WL(-Z)	228.1	0	0	1.2	2.5	-103.1	0	0	64	131.2
227 0.6(DL+EQDL)+0.7SL(+X)	-1.8	0	0	1.2	2.5	35.7	0	0	1.4	2.9
228 0.6(DL+EQDL)+0.7SL(-X)	120	0	0	1.2	2.5	128.1	0	0	-1.4	-2.9
229 0.6(DL+EQDL)+0.7SL(+Z)	-96.3	0	0	2.6	5.3	254.8	0	0	-44.1	-90.4
230 0.6(DL+EQDL)+0.7SL(-Z)	214.5	0	0	-0.2	-0.4	-91	0	0	44.1	90.4

Node	3					4				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL+EQDL+FL	229.5	-3	6.2	1	2.1	440.5	4	-8.2	-7	-14.4
202 DL+EQDL+FL+LL+EQLL+TL	789.5	-3	6.2	12	24.6	978.5	4	-8.2	-26	-53.3
203 DL+EQDL+FL+0.75(LL+EQLL)+	649.5	-3	6.2	9.3	19	844	4	-8.2	-21.3	-43.6
204 DL+EQDL+FL+PL	77.5	-27	55.4	1	2.1	416.5	-25	51.3	-36	-73.8
205 DL+EQDL+FL+PL+LL+EQLL+T	637.5	-27	55.4	12	24.6	954.5	-25	51.3	-55	-112.8
206 DL+EQDL+FL+PL+0.75(LL+EQ	497.5	-27	55.4	9.3	19	820	-25	51.3	-50.3	-103
207 DL+EQDL+FL+WL(+X)	9.5	-69	141.5	1	2.1	198.5	-58	118.9	-13	-26.7
208 DL+EQDL+FL+WL(-X)	449.5	63	-129.2	1	2.1	682.5	66	-135.3	-1	-2
209 DL+EQDL+FL+WL(+Z)	7.5	2	-4.1	1	2.1	636.5	-1	2.1	-93	-190.7
210 DL+EQDL+FL+WL(-Z)	451.5	-8	16.4	1	2.1	244.5	9	-18.5	79	162
211 DL+EQDL+FL+0.7SL(+X)	111.2	-36.6	75	1	2.1	292.8	-29.6	60.7	-13.3	-27.3
212 DL+EQDL+FL+0.7SL(-X)	347.8	30.6	-62.7	1	2.1	588.2	37.6	-77.1	-0.7	-1.4
213 DL+EQDL+FL+0.7SL(+Z)	37.7	3.3	-6.8	3.1	6.4	600.8	-2.3	4.7	-63.7	-130.6
214 DL+EQDL+FL+0.7SL(-Z)	421.3	-9.3	19.1	-1.1	-2.3	280.2	10.3	-21.1	49.7	101.9

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215 DL+EQDL+FL+0.75WL(+X)+0.7	489.8	-52.5	107.6	7	14.4	655.8	-42.5	87.1	-21.3	-43.6
216 DL+EQDL+FL+0.75WL(-X)+0.7	819.8	46.5	-95.3	7	14.4	1018.8	50.5	-103.5	-12.3	-25.1
217 DL+EQDL+FL+0.75WL(+Z)+0.7	488.3	0.8	-1.5	7	14.4	984.3	0.2	-0.5	-81.3	-166.6
218 DL+EQDL+FL+0.75WL(-Z)+0.7	821.3	-6.8	13.8	7	14.4	690.3	7.8	-15.9	47.8	97.9
219 DL+EQDL+FL+0.525SL(+X)+0.	566	-28.2	57.8	7	14.4	726.5	-21.2	43.5	-21.5	-44
220 DL+EQDL+FL+0.525SL(-X)+0.7	743.5	22.2	-45.5	7	14.4	948	29.2	-59.9	-12	-24.7
221 DL+EQDL+FL+0.525SL(+Z)+0.	510.9	1.7	-3.5	8.6	17.6	957.5	-0.7	1.5	-59.3	-121.5
222 DL+EQDL+FL+0.525SL(-Z)+0.7	798.6	-7.7	15.8	5.4	11.1	717	8.7	-17.9	25.8	52.8
223 0.6(DL+EQDL)+WL(+X)	-103.9	-64.2	131.6	1.8	3.7	-78.5	-62	127.1	-7.8	-16
224 0.6(DL+EQDL)+WL(-X)	336.1	67.8	-139	1.8	3.7	405.5	62	-127.1	4.2	8.6
225 0.6(DL+EQDL)+WL(+Z)	-105.9	6.8	-13.9	1.8	3.7	359.5	-5	10.3	-87.8	-180
226 0.6(DL+EQDL)+WL(-Z)	338.1	-3.2	6.6	1.8	3.7	-32.5	5	-10.3	84.2	172.6
227 0.6(DL+EQDL)+0.7SL(+X)	-2.2	-31.8	65.2	1.8	3.7	15.8	-33.6	68.9	-8.1	-16.6
228 0.6(DL+EQDL)+0.7SL(-X)	234.4	35.4	-72.6	1.8	3.7	311.2	33.6	-68.9	4.5	9.2
229 0.6(DL+EQDL)+0.7SL(+Z)	-75.7	8.1	-16.6	3.9	8	323.8	-6.3	12.9	-58.5	-119.9
230 0.6(DL+EQDL)+0.7SL(-Z)	307.9	-4.5	9.2	-0.3	-0.6	3.2	6.3	-12.9	54.9	112.5

Node	5					6				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL+EQDL+FL	160.5	-4	8.2	10	20.5	379.5	3	-6.2	-3	-6.2
202 DL+EQDL+FL+LL+EQLL+TL	545.5	-6	12.3	36	73.8	706.5	3	-6.2	-25	-51.3
203 DL+EQDL+FL+0.75(LL+EQLL)+	449.3	-5.5	11.3	29.5	60.5	624.8	3	-6.2	-19.5	-40
204 DL+EQDL+FL+PL	152.5	-28	57.4	-17	-34.9	581.5	-26	53.3	-30	-61.5
205 DL+EQDL+FL+PL+LL+EQLL+T	537.5	-30	61.5	9	18.5	908.5	-26	53.3	-52	-106.6
206 DL+EQDL+FL+PL+0.75(LL+EQ	441.3	-29.5	60.5	2.5	5.1	826.8	-26	53.3	-46.5	-95.3
207 DL+EQDL+FL+WL(+X)	437.5	-70	143.5	13	26.7	658.5	-59	121	-1	-2.1
208 DL+EQDL+FL+WL(-X)	-116.5	62	-127.1	7	14.4	100.5	65	-133.3	-5	-10.3
209 DL+EQDL+FL+WL(+Z)	-9.5	1	-2.1	-19	-39	560.5	-2	4.1	-32	-65.6
210 DL+EQDL+FL+WL(-Z)	330.5	-9	18.5	39	80	198.5	8	-16.4	26	53.3
211 DL+EQDL+FL+0.7SL(+X)	339	-37.6	77.1	12.8	26.2	572.7	-30.6	62.7	-0.2	-0.4
212 DL+EQDL+FL+0.7SL(-X)	-18	29.6	-60.7	7.2	14.8	186.3	36.6	-75	-5.8	-11.9
213 DL+EQDL+FL+0.7SL(+Z)	17	2.3	-4.7	-8.2	-16.8	536.3	-3.3	6.8	-21.2	-43.5
214 DL+EQDL+FL+0.7SL(-Z)	304	-10.3	21.1	28.2	57.8	222.7	9.3	-19.1	15.2	31.2
215 DL+EQDL+FL+0.75WL(+X)+0.7	651.8	-53.5	109.7	28	57.4	840	-43.5	89.2	-15.8	-32.3
216 DL+EQDL+FL+0.75WL(-X)+0.7	236.3	45.5	-93.3	23.5	48.2	421.5	49.5	-101.5	-18.8	-38.4
217 DL+EQDL+FL+0.75WL(+Z)+0.7	316.5	-0.2	0.5	4	8.2	766.5	-0.8	1.5	-39	-80
218 DL+EQDL+FL+0.75WL(-Z)+0.7	571.5	-7.8	15.9	47.5	97.4	495	6.8	-13.8	4.5	9.2
219 DL+EQDL+FL+0.525SL(+X)+0.	577.9	-29.2	59.9	27.9	57.1	775.7	-22.2	45.5	-15.2	-31.1
220 DL+EQDL+FL+0.525SL(-X)+0.7	310.1	21.2	-43.5	23.7	48.5	485.9	28.2	-57.8	-19.4	-39.7
221 DL+EQDL+FL+0.525SL(+Z)+0.	336.4	0.7	-1.5	12.1	24.8	748.4	-1.7	3.5	-30.9	-63.3
222 DL+EQDL+FL+0.525SL(-Z)+0.7	551.6	-8.7	17.9	39.4	80.8	513.2	7.7	-15.8	-3.6	-7.4
223 0.6(DL+EQDL)+WL(+X)	350.5	-66	135.3	6.6	13.5	388.5	-63.8	130.8	-2.2	-4.5
224 0.6(DL+EQDL)+WL(-X)	-203.5	66	-135.3	0.6	1.2	-169.5	60.2	-123.4	-6.2	-12.7
225 0.6(DL+EQDL)+WL(+Z)	-96.5	5	-10.3	-25.4	-52.1	290.5	-6.8	13.9	-33.2	-68.1
226 0.6(DL+EQDL)+WL(-Z)	243.5	-5	10.3	32.6	66.8	-71.5	3.2	-6.6	24.8	50.8
227 0.6(DL+EQDL)+0.7SL(+X)	252	-33.6	68.9	6.4	13.1	302.7	-35.4	72.6	-1.4	-2.9
228 0.6(DL+EQDL)+0.7SL(-X)	-105	33.6	-68.9	0.8	1.6	-83.7	31.8	-65.2	-7	-14.4
229 0.6(DL+EQDL)+0.7SL(+Z)	-70	6.3	-12.9	-14.6	-29.9	266.3	-8.1	16.6	-22.4	-45.9
230 0.6(DL+EQDL)+0.7SL(-Z)	217	-6.3	12.9	21.8	44.7	-47.3	4.5	-9.2	14	28.7

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	Load at top of foundation					Load at bottom of foundation				
L/C	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 DL+EQDL+FL	1461.0	0.0	-1768.8	1.0	-2794.9	4530.6	0.0	-1768.8	1.0	-2794.9
202 DL+EQDL+FL+LL+EQLL+TL	3783.0	-2.0	-3071.2	3.0	-4520.6	6852.6	-2.0	-3071.2	3.0	-4520.6
203 DL+EQDL+FL+0.75(LL+EQLL)+	3202.6	-1.5	-2746.1	2.5	-4089.2	6272.2	-1.5	-2746.1	2.5	-4089.2
204 DL+EQDL+FL+PL	1460.0	-106.0	-2905.8	-106.0	-4378.3	4529.6	-106.0	-2905.8	-106.0	-4378.3
205 DL+EQDL+FL+PL+LL+EQLL+T	3782.0	-108.0	-4208.2	-104.0	-6104.0	6851.6	-108.0	-4208.2	-104.0	-6104.0
206 DL+EQDL+FL+PL+0.75(LL+EQ	3201.6	-107.5	-3883.2	-104.5	-5672.6	6271.2	-107.5	-3883.2	-104.5	-5672.6
207 DL+EQDL+FL+WL(+X)	1460.0	-256.0	-5323.7	2.0	-2788.9	4529.6	-256.0	-5323.7	2.0	-2788.9
208 DL+EQDL+FL+WL(-X)	1462.0	256.0	232.5	0.0	-3199.5	4531.6	256.0	232.5	0.0	-3199.5
209 DL+EQDL+FL+WL(+Z)	1462.0	0.0	-2514.6	-207.0	-6542.8	4531.6	0.0	-2514.6	-207.0	-6542.8
210 DL+EQDL+FL+WL(-Z)	1460.0	0.0	-2037.5	209.0	774.1	4529.6	0.0	-2037.5	209.0	774.1
211 DL+EQDL+FL+0.7SL(+X)	1459.6	-134.4	-4434.6	1.7	-2792.1	4529.2	-134.4	-4434.6	1.7	-2792.1
212 DL+EQDL+FL+0.7SL(-X)	1462.4	134.4	652.8	0.3	-2867.3	4532.0	134.4	652.8	0.3	-2867.3
213 DL+EQDL+FL+0.7SL(+Z)	1460.3	0.0	-2333.3	-132.7	-6007.0	4529.9	0.0	-2333.3	-132.7	-6007.0
214 DL+EQDL+FL+0.7SL(-Z)	1461.7	0.0	-1916.8	134.7	300.0	4531.3	0.0	-1916.8	134.7	300.0
215 DL+EQDL+FL+0.75WL(+X)+0.7	3201.2	-192.0	-5420.1	2.4	-4083.1	6270.8	-192.0	-5420.1	2.4	-4083.1
216 DL+EQDL+FL+0.75WL(-X)+0.7	3202.7	192.0	-87.5	0.9	-4091.9	6272.3	192.0	-87.5	0.9	-4091.9
217 DL+EQDL+FL+0.75WL(+Z)+0.7	3202.7	0.0	-2747.8	-154.3	-6935.1	6272.3	0.0	-2747.8	-154.3	-6935.1
218 DL+EQDL+FL+0.75WL(-Z)+0.7	3201.2	0.0	-2758.6	157.9	-1240.6	6270.8	0.0	-2758.6	157.9	-1240.6
219 DL+EQDL+FL+0.525SL(+X)+0.	3200.8	-100.8	-4753.4	2.3	-4086.2	6270.4	-100.8	-4753.4	2.3	-4086.2
220 DL+EQDL+FL+0.525SL(-X)+0.7	3202.9	100.8	-754.2	1.3	-4090.0	6272.5	100.8	-754.2	1.3	-4090.0
221 DL+EQDL+FL+0.525SL(+Z)+0.	3201.4	0.0	-2754.5	-98.5	-6499.6	6271.0	0.0	-2754.5	-98.5	-6499.6
222 DL+EQDL+FL+0.525SL(-Z)+0.7	3202.3	0.0	-2753.1	102.0	-1676.1	6271.9	0.0	-2753.1	102.0	-1676.1
223 0.6(DL+EQDL)+WL(+X)	602.6	-256.0	-3593.1	1.6	-1417.3	3672.2	-256.0	-3593.1	1.6	-1417.3
224 0.6(DL+EQDL)+WL(-X)	604.6	256.0	-880.9	-0.4	-1690.8	3674.2	256.0	-880.9	-0.4	-1690.8
225 0.6(DL+EQDL)+WL(+Z)	604.6	0.0	-2027.2	-207.4	-3704.8	3674.2	0.0	-2027.2	-207.4	-3704.8
226 0.6(DL+EQDL)+WL(-Z)	602.6	0.0	-1684.2	208.6	956.6	3672.2	0.0	-1684.2	208.6	956.6
227 0.6(DL+EQDL)+0.7SL(+X)	602.2	-134.4	-2959.3	1.3	-877.4	3671.8	-134.4	-2959.3	1.3	-877.4
228 0.6(DL+EQDL)+0.7SL(-X)	605.0	134.4	-70.5	-0.1	-1438.3	3674.6	134.4	-70.5	-0.1	-1438.3
229 0.6(DL+EQDL)+0.7SL(+Z)	602.9	0.0	-1811.0	-133.1	-3429.5	3672.5	0.0	-1811.0	-133.1	-3429.5
230 0.6(DL+EQDL)+0.7SL(-Z)	604.3	0.0	-1363.4	134.3	961.9	3673.9	0.0	-1363.4	134.3	961.9

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L/C	Load at b/o fdn					F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
	V (kN)	Hx (kN)	Mz (kNm)	Hz (kN)	Mx (kNm)				
201 DL+EQDL+FL	4530.6	0.0	1768.8	1.0	2794.9	15.8	6.3	NA	1812.2
202 DL+EQDL+FL+LL+EQLL+TI	6852.6	-2.0	3071.2	3.0	4520.6	13.7	5.9	1370.5	913.7
203 DL+EQDL+FL+0.75(LL+EQ	6272.2	-1.5	2746.1	2.5	4089.2	14.0	6.0	1672.6	1003.5
204 DL+EQDL+FL+PL	4529.6	-106.0	2905.8	-106.0	4378.3	9.6	4.0	17.1	17.1
205 DL+EQDL+FL+PL+LL+EQLI	6851.6	-108.0	4208.2	-104.0	6104.0	10.0	4.4	25.4	26.4
206 DL+EQDL+FL+PL+0.75(LL+	6271.2	-107.5	3883.2	-104.5	5672.6	9.9	4.3	23.3	24.0
207 DL+EQDL+FL+WL(+X)	4529.6	-256.0	5323.7	2.0	2788.9	5.2	6.3	7.1	905.9
208 DL+EQDL+FL+WL(-X)	4531.6	256.0	232.5	0.0	3199.5	119.8	5.5	7.1	NA
209 DL+EQDL+FL+WL(+Z)	4531.6	0.0	2514.6	-207.0	6542.8	11.1	2.7	NA	8.8
210 DL+EQDL+FL+WL(-Z)	4529.6	0.0	2037.5	209.0	774.1	13.7	22.8	NA	8.7
211 DL+EQDL+FL+0.7SL(+X)	4529.2	-134.4	4434.6	1.7	2792.1	6.3	6.3	13.5	1065.7
212 DL+EQDL+FL+0.7SL(-X)	4532.0	134.4	652.8	0.3	2867.3	42.7	6.2	13.5	6042.6
213 DL+EQDL+FL+0.7SL(+Z)	4529.9	0.0	2333.3	-132.7	6007.0	11.9	2.9	NA	13.7
214 DL+EQDL+FL+0.7SL(-Z)	4531.3	0.0	1916.8	134.7	300.0	14.5	58.9	NA	13.5
215 DL+EQDL+FL+0.75WL(+X)+	6270.8	-192.0	5420.1	2.4	4083.1	7.1	6.0	13.1	1045.1
216 DL+EQDL+FL+0.75WL(-X)+	6272.3	192.0	87.5	0.9	4091.9	440.9	6.0	13.1	2787.7
217 DL+EQDL+FL+0.75WL(+Z)+	6272.3	0.0	2747.8	-154.3	6935.1	14.0	3.5	NA	16.3
218 DL+EQDL+FL+0.75WL(-Z)+	6270.8	0.0	2758.6	157.9	1240.6	14.0	19.7	NA	15.9
219 DL+EQDL+FL+0.525SL(+X)	6270.4	-100.8	4753.4	2.3	4086.2	8.1	6.0	24.9	1090.5
220 DL+EQDL+FL+0.525SL(-X)+	6272.5	100.8	754.2	1.3	4090.0	51.1	6.0	24.9	1930.0
221 DL+EQDL+FL+0.525SL(+Z)	6271.0	0.0	2754.5	-98.5	6499.6	14.0	3.8	NA	25.5
222 DL+EQDL+FL+0.525SL(-Z)+	6271.9	0.0	2753.1	102.0	1676.1	14.0	14.6	NA	24.6
223 0.6(DL+EQDL)+WL(+X)	3672.2	-256.0	3593.1	1.6	1417.3	6.3	10.1	5.7	918.0
224 0.6(DL+EQDL)+WL(-X)	3674.2	256.0	880.9	-0.4	1690.8	25.7	8.5	5.7	3674.2
225 0.6(DL+EQDL)+WL(+Z)	3674.2	0.0	2027.2	-207.4	3704.8	11.1	3.9	NA	7.1
226 0.6(DL+EQDL)+WL(-Z)	3672.2	0.0	1684.2	208.6	956.6	13.4	15.0	NA	7.0
227 0.6(DL+EQDL)+0.7SL(+X)	3671.8	-134.4	2959.3	1.3	877.4	7.6	16.3	10.9	1129.8
					Min.=	**5.2**	**2.7**	**5.7**	**7.0**

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Base pressure check of footing F1 :

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	1	201	0	45.316	0
Min Px	1	201	0	45.316	0
Max Py	1812	205	0	227.619	0
Min Py	1801	208	0	0	0
Max Pz	1	201	0	45.316	0
Min Pz	1	201	0	45.316	0

Gross Maximum Base pressure, p_{max} = 227.6 KN/m^2

Net Allowable Safe Bearing Capacity = 450 KN/m^2
Depth of soil above footing = 0.75 m
Unit weight of soil = 18 KN/m^3

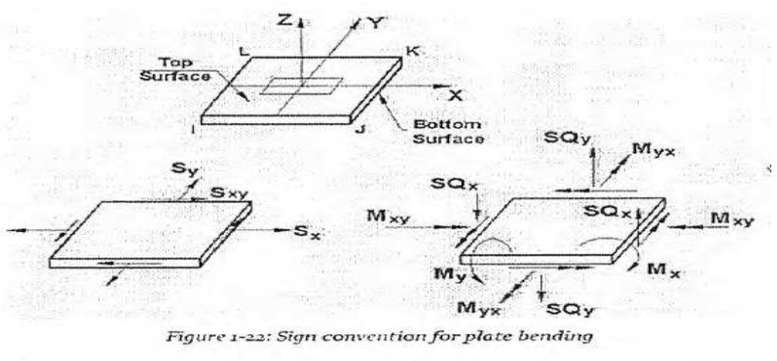
Gross Safe Bearing Capacity = 463.5 KN/m^2
> 227.6 KN/m^2
Hence OK

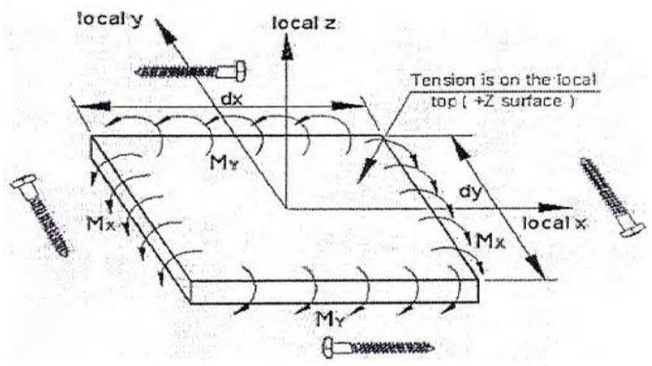
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Design of Foundation Raft F1 :





Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; : FDN_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	4514	0.757	-0.06	0	0	0	-187.5	-107.944	11.996	-199.496	-119.94	106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)
Min Qx	4432	-0.676	-0.043	0	0	0	-185.4	-116.063	0	-185.449	-116.063	103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
Max Qy	4250	0.049	1.186	0	0	0	-4.819	-41.302	-36.309	-41.128	-77.611	105 1.2(DL+EQDL+FL+PL)+1.6(LL+EQLL)
Min Qy	4334	0.059	-1.047	0	0	0	-0.834	-60.459	54.9	-55.734	-115.359	111 1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)
Max Sx	5865	-0.114	-0.05	0	0	0	21.744	-10.911	22.505	44.249	-33.416	101 1.4(DL+EQDL+FL)
Min Sx	5865	-0.114	-0.05	0	0	0	21.744	-10.911	22.505	44.249	-33.416	101 1.4(DL+EQDL+FL)
Max Sy	5865	-0.114	-0.05	0	0	0	21.744	-10.911	22.505	44.249	-33.416	101 1.4(DL+EQDL+FL)
Min Sy	5865	-0.114	-0.05	0	0	0	21.744	-10.911	22.505	44.249	-33.416	101 1.4(DL+EQDL+FL)
Max Sxy	5865	-0.114	-0.05	0	0	0	21.744	-10.911	22.505	44.249	-33.416	101 1.4(DL+EQDL+FL)
Min Sxy	5865	-0.114	-0.05	0	0	0	21.744	-10.911	22.505	44.249	-33.416	101 1.4(DL+EQDL+FL)
Max Mx	4467	-0.19	0.021	0	0	0	286.72	4.345	-29.056	315.778	33.401	105 1.2(DL+EQDL+FL+PL)+1.6(LL+EQLL)
Min Mx	4396	-0.112	-0.489	0	0	0	-269.8	-67.607	11.379	-281.216	-78.986	103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
Max My	4284	-0.05	0.085	0	0	0	18.891	276.752	-42.289	61.18	319.041	106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)
Min My	4135	0.269	0.244	0	0	0	-17.12	-190.748	-37.922	-55.042	-228.67	105 1.2(DL+EQDL+FL+PL)+1.6(LL+EQLL)
Max Mxy	5839	-0.201	-0.143	0	0	0	116.67	9.034	145.103	261.771	154.137	111 1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)
Min Mxy	5293	-0.369	0.359	0	0	0	80.98	9.969	-191.007	271.987	200.976	105 1.2(DL+EQDL+FL+PL)+1.6(LL+EQLL)

PLATE CENTER STRESS ; FDN_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	4441	0.406	-0.024	0	0	0	-45.29	-7.646	9.354	-54.641	-17	106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)
Min Qx	4479	-0.533	-0.018	0	0	0	171.69	-6.78	0	171.685	-6.78	105 1.2(DL+EQDL+FL+PL)+1.6(LL+EQLL)
Max Qy	4256	-0.075	0.516	0	0	0	11.388	132.203	-125.751	137.139	257.954	105 1.2(DL+EQDL+FL+PL)+1.6(LL+EQLL)
Min Qy	4324	-0.094	-0.4	0	0	0	19.677	82.586	98.725	118.402	181.311	111 1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)
Max Sx	5863	-0.053	-0.03	0	0	0	40.378	6.697	16.311	56.689	23.008	101 1.4(DL+EQDL+FL)
Min Sx	5863	-0.053	-0.03	0	0	0	40.378	6.697	16.311	56.689	23.008	101 1.4(DL+EQDL+FL)
Max Sy	5863	-0.053	-0.03	0	0	0	40.378	6.697	16.311	56.689	23.008	101 1.4(DL+EQDL+FL)
Min Sy	5863	-0.053	-0.03	0	0	0	40.378	6.697	16.311	56.689	23.008	101 1.4(DL+EQDL+FL)
Max Sxy	5863	-0.053	-0.03	0	0	0	40.378	6.697	16.311	56.689	23.008	101 1.4(DL+EQDL+FL)
Min Sxy	5863	-0.053	-0.03	0	0	0	40.378	6.697	16.311	56.689	23.008	101 1.4(DL+EQDL+FL)
Max Mx	4467	-0.19	0.021	0	0	0	286.72	4.345	-29.056	315.778	33.401	105 1.2(DL+EQDL+FL+PL)+1.6(LL+EQLL)
Min Mx	4861	-0.012	0.242	0	0	0	-122	40.443	7.721	-129.745	48.164	112 1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)
Max My	4284	-0.05	0.085	0	0	0	18.891	276.752	-42.289	61.18	319.041	106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)
Min My	4292	0.021	0.012	0	0	0	-9.23	-78.786	-6.492	-15.722	-85.278	120 0.9(DL+EQDL)+1.6WL(-X)
Max Mxy	5788	-0.118	-0.223	0	0	0	47.51	81.425	115.134	162.644	196.559	111 1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)
Min Mxy	5345	-0.143	0.232	0	0	0	72.406	123.181	-158.815	231.221	281.996	105 1.2(DL+EQDL+FL+PL)+1.6(LL+EQLL)

Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		750	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

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Reinforcement Along X:

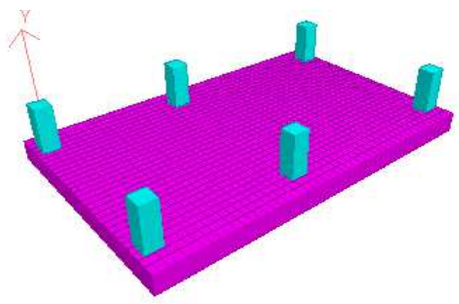
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c\ *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	315.78	top face	50	692	315.78	0.88	0.0021	1470.7	1350	16	136.76	125	1609	22.69	28.36	0.07	tension control
Min Mx	-281.22	bottom face	50	692	281.22	0.78	0.0019	1307.5		16	148.99	125	1609	22.69	28.36	0.07	tension control
Max Mxy	261.77	top face	50	692	261.77	0.73	0.0018	1216.0		16	148.99	125	1609	22.69	28.36	0.07	tension control
Min Mxy	271.99	top face	50	692	271.99	0.76	0.0018	1264.1		16	148.99	125	1609	22.69	28.36	0.07	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.406	1	304.5	54.641	0.002	0.692	760.524	682.385	1187.24	682.38	2538	> 304.5	ok	511.8	φVc>Vu	not required
	Min Qx	0.533	1	399.75	171.69			699.106			682.38	2538	> 399.75	ok	511.8	φVc>Vu	not required

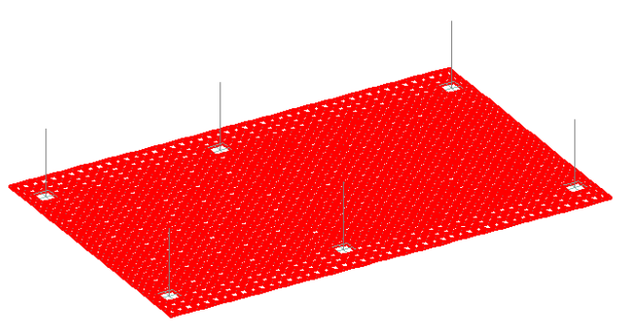
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c\ *(d-c)$	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	319.04	top face	50	692	319.04	0.89	0.0021	1486.2	1350	16	135.34	125	1609.2	22.69	28.36	0.07	tension control
Min My	-228.67	bottom face	50	692	228.67	0.64	0.0015	1060.5		16	148.99	125	1609.2	22.69	28.36	0.07	tension control
Max Mxy	154.14	top face	50	692	154.14	0.43	0.0010	712.3		16	148.99	125	1609.2	22.69	28.36	0.07	tension control
Min Mxy	200.98	top face	50	692	200.98	0.56	0.0013	930.8		16	148.99	125	1609.2	22.69	28.36	0.07	tension control

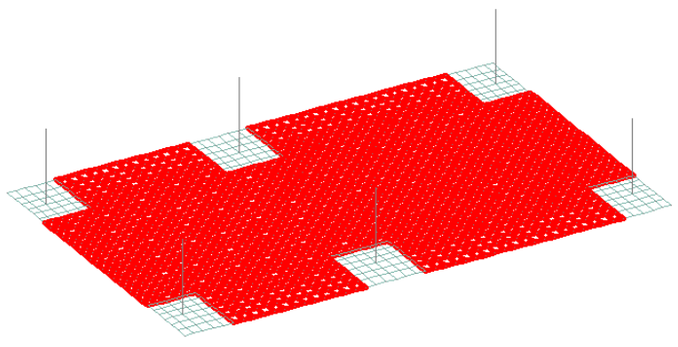
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c + 0.66\sqrt{f_c'}bd) \geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.516	1	387	257.95	0.002	0.692	683.429	fc'682.4	fc'1187	682.38	714	> 387	ok	511.8	φVc>Vu	not required
	Min Qy	0.4	1	300	181.31			686.351			682.38	714	> 300	ok	511.8	φVc>Vu	not required



3D Rendered View of foundation



Highlighted plate for moment (at column face)



Highlighted plate for shear (at d distance from column face)

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
- At Bottom **16ϕT** **125** c/c
- At Top **16ϕT** **125** c/c

Along Z Direction :
- At Bottom **16ϕT** **125** c/c
- At Top **16ϕT** **125** c/c

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5.2 DESIGN OF COLUMN C1

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.65

% of reinforcement, p_t = 1.5 % i.e. ρ_g = 0.015

β = 0.65

k = 1.2

Column Dimensions :

Depth parallel to X-X axis, h = 550 mm

Depth parallel to Z-Z axis, b = 550 mm

Clear Cover = 50 mm

d' = 70 mm

Y_h = (h-d')/h = 0.75

Y_b = (b-d')/b = 0.75

Unsupported length of column parallel to X-X axis, l_{ux} = 1.75 m kl_{ux}/r_x = 12.73 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 1.75 m kl_{uz}/r_z = 12.73

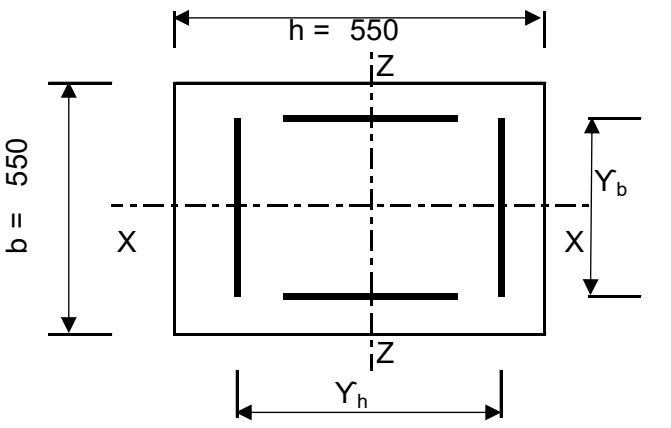
Radius of gyration r_x= 165 mm

$P_{n, Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 8769.55 KN safe

Radius of gyration r_z= 165 mm

L/C	Factored Force Value P_u (KN)	M_{ux} (KN-m)	M_{uz} (KN-m)	Design force Value $P_n = P_u/φ$ (KN)	$M_{nx}=M_{ux}/φ$ (KN-m)	$M_{nz}=M_{uz}/φ$ (KN-m)	$K_n = P_n/f_c'bh$	Uniaxial moment capacity about X-X axis: $R_{nx}= M_{nx1}/f_c'b^2h$	M_{nx1} (KN-m)	M_{nx}/M_{nx1}	about ZZ axis: $R_{nz}= M_{nz1}/f_c'h^2b$	M_{nz1} (KN-m)	M_{nz}/M_{nz1}	Check for Interaction Ratio: α=log0.5/logβ	$(M_{nx}/M_{nx1})^α + (M_{nz}/M_{nz1})^α$	Remarks
103	1382.4	61.0	8.0	2127	93.8	12.3	0.2009	0.090	524.081	0.18	0.090	524.1	0.02	1.609	0.066	safe
120	346.9	0.0	0.0	534	0.0	0.0	0.0504	0.070	407.619	0.00	0.070	407.6	0.00		0	safe
111	775.8	37.7	176.9	1194	58.0	272.2	0.1127	0.075	436.734	0.13	0.075	436.7	0.62		0.507	safe
112	864.3	12.7	182.9	1330	19.5	281.4	0.1256	0.075	436.734	0.04	0.075	436.7	0.64		0.5	safe
122	70.9	226.0	13.4	109	347.7	20.6	0.0103	0.070	407.619	0.85	0.070	407.6	0.05		0.783	safe
113	901.7	247.4	13.4	1387	380.6	20.6	0.1310	0.075	436.734	0.87	0.075	436.7	0.05		0.809	safe
101	126.9	0.0	0.0	195	0.0	0.0	0.0184	0.070	407.619	0.00	0.070	407.6	0.00		0	safe
101	126.9	0.0	0.0	195	0.0	0.0	0.0184	0.070	407.619	0.00	0.070	407.6	0.00		0	safe
113	901.7	247.4	13.4	1387	380.6	20.6	0.1310	0.075	436.734	0.87	0.075	436.7	0.05		0.809	safe
122	70.9	226.0	13.4	109	347.7	20.6	0.0103	0.070	407.619	0.85	0.070	407.6	0.05		0.783	safe
111	775.8	37.7	176.9	1194	58.0	272.2	0.1127	0.075	436.734	0.13	0.075	436.7	0.62		0.507	safe
112	864.3	12.7	182.9	1330	19.5	281.4	0.1256	0.075	436.734	0.04	0.075	436.7	0.64		0.5	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
4538	4	20	+	12	20	5029	1.66	10	320	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

ACI DESIGN HANDBOOK-SP-17M(09)

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST- CRT-1001) Clause 4.1

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ATTACHMENT 1 STAAD INPUT & OUTPUT FILE FOR F1 DESIGN

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*           Bentley Systems, Inc.                 *
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71. 2012 3.86842 0 6.56667; 2013 4.12632 0 6.9; 2014 4.12632 0 6.56667
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904. 5783 3262 3263 3288 3287; 5784 3263 3264 3289 3288; 5785 3264 3265 3290 3289
905. 5786 3265 3266 3291 3290; 5787 3266 3267 3292 3291; 5788 3267 1972 1975 3292
906. 5789 2729 3268 3293 2748; 5790 3268 3269 3294 3293; 5791 3269 3270 3295 3294
907. 5792 3270 3271 3296 3295; 5793 3271 3272 3297 3296; 5794 3272 3273 3298 3297
908. 5795 3273 3274 3299 3298; 5796 3274 3275 3300 3299; 5797 3275 3276 3301 3300
909. 5798 3276 3277 3302 3301; 5799 3277 3278 3303 3302; 5800 3278 3279 3304 3303
910. 5801 3279 3280 3305 3304; 5802 3280 3281 3306 3305; 5803 3281 3282 3307 3306
911. 5804 3282 3283 3308 3307; 5805 3283 3284 3309 3308; 5806 3284 3285 3310 3309
912. 5807 3285 3286 3311 3310; 5808 3286 3287 3312 3311; 5809 3287 3288 3313 3312
913. 5810 3288 3289 3314 3313; 5811 3289 3290 3315 3314; 5812 3290 3291 3316 3315
914. 5813 3291 3292 3317 3316; 5814 3292 1975 1978 3317; 5815 2748 3293 3318 2767
915. 5816 3293 3294 3319 3318; 5817 3294 3295 3320 3319; 5818 3295 3296 3321 3320
916. 5819 3296 3297 3322 3321; 5820 3297 3298 3323 3322; 5821 3298 3299 3324 3323
917. 5822 3299 3300 3325 3324; 5823 3300 3301 3326 3325; 5824 3301 3302 3327 3326
918. 5825 3302 3303 3328 3327; 5826 3303 3304 3329 3328; 5827 3304 3305 3330 3329
919. 5828 3305 3306 3331 3330; 5829 3306 3307 3332 3331; 5830 3307 3308 3333 3332
920. 5831 3308 3309 3334 3333; 5832 3309 3310 3335 3334; 5833 3310 3311 3336 3335
921. 5834 3311 3312 3337 3336; 5835 3312 3313 3338 3337; 5836 3313 3314 3339 3338
922. 5837 3314 3315 3340 3339; 5838 3315 3316 3341 3340; 5839 3316 3317 3342 3341
923. 5840 3317 1978 1981 3342; 5841 2767 3318 2231 3; 5842 3318 3319 2233 2231
924. 5843 3319 3320 2235 2233; 5844 3320 3321 2237 2235; 5845 3321 3322 2239 2237
925. 5846 3322 3323 2241 2239; 5847 3323 3324 2243 2241; 5848 3324 3325 2245 2243
926. 5849 3325 3326 2247 2245; 5850 3326 3327 2249 2247; 5851 3327 3328 2251 2249
927. 5852 3328 3329 2253 2251; 5853 3329 3330 2255 2253; 5854 3330 3331 2257 2255
928. 5855 3331 3332 2259 2257; 5856 3332 3333 2261 2259; 5857 3333 3334 2263 2261
929. 5858 3334 3335 2265 2263; 5859 3335 3336 2267 2265; 5860 3336 3337 2269 2267
930. 5861 3337 3338 2271 2269; 5862 3338 3339 2273 2271; 5863 3339 3340 2275 2273
931. 5864 3340 3341 2277 2275; 5865 3341 3342 2279 2277; 5866 3342 1981 5 2279
932. START GROUP DEFINITION
933. ELEMENT
934. _RAFT 4102 4104 4106 4108 4110 4111 4114 4116 4118 4120 4122 4123 4126 4128 -

935. 4130 4132 4134 4135 4138 4140 4142 4144 4146 4147 4150 4152 4154 4156 4158 -
936. 4160 4162 4164 4166 4168 4170 4172 4174 4176 4178 4180 4182 4184 4186 4188 -
937. 4190 4192 4194 4196 4198 4200 4202 4204 4206 4208 4210 4212 4214 4216 4218 -
938. 4220 4222 4224 4226 4228 4230 4232 4234 4236 4238 4240 TO 4242 4244 4246 -
939. 4248 4250 4252 4254 4256 4258 4260 4262 4264 4266 4268 4270 4272 4274 4276 -
940. 4278 4280 4282 4284 4286 4288 4290 4292 4294 4296 4298 4300 4302 4304 4306 -
941. 4308 4310 4312 4314 4316 4318 4320 4322 4324 4326 4328 4330 4332 -
942. 4334 TO 4336 4340 4342 4344 4346 4348 4350 4352 4354 4356 4358 4360 4362 -
943. 4364 4366 4368 4370 4372 4374 4376 TO 4394 4396 4398 4400 4402 4404 4406 -
944. 4408 4410 4412 4414 4416 4418 4420 4422 4424 4426 4428 4430 4432 4433 4435 -
945. 4437 4439 4441 4443 4445 4447 4449 4451 4453 4455 4457 4459 4461 4463 4465 -
946. 4467 4469 4471 4473 4475 4477 4479 4481 4483 TO 4510 4512 4514 4516 4518 -
947. 4520 4522 4524 4526 4528 4530 4532 4534 4536 4538 4540 4542 4544 4546 4548 -
948. 4550 4552 4554 4556 4558 4560 4561 4563 4565 4567 4569 4571 4573 4575 4577 -
949. 4579 4581 4583 4585 4587 4589 4591 4593 4595 4597 4599 TO 4617 4619 4621 -
950. 4623 4625 4627 4629 4631 4633 4635 4637 4639 4641 4643 4645 4647 4649 4651 -
951. 4653 4655 4656 4658 4660 4662 4664 4666 4668 4670 4672 4674 4676 4678 4680 -
952. 4682 4684 4686 4688 4690 4692 4694 4696 4698 4700 4702 4704 4706

WARNING- One or more lines are too long and will be split into 2 lines.
This may not work for all commands. Please check.

953. _RAFT 4707 TO 4733 4735 4737 4739 4741 4743 4745 4747 4749 4751 4753 4755 -
954. 4757 -
955. 4759 4761 4763 4765 4767 4769 4771 4773 4775 4777 4779 4781 4783 4784 4786 -
956. 4787 TO 5241 5243 TO 5866
957. _RAFT_MOM 4102 4104 4106 4108 4110 4114 4118 4120 4122 4123 4126 4128 4130 -
958. 4132 4135 4140 4142 4144 4146 4147 4150 4154 4156 4158 4160 4162 4164 4166 -
959. 4168 4170 4172 4174 4176 4178 4180 4182 4184 4186 4188 4190 4192 4194 4196 -
960. 4198 4200 4202 4204 4206 4208 4210 4212 4214 4216 4218 4220 4222 4224 4226 -
961. 4228 4230 4232 4234 4236 4238 4240 4241 4246 4248 4250 4252 4254 4256 4258 -
962. 4260 4262 4264 4266 4268 4270 4272 4274 4276 4278 4280 4282 4284 4286 4288 -
963. 4290 4292 4294 4296 4298 4300 4302 4304 4306 4308 4310 4312 4314 4316 4318 -
964. 4320 4322 4324 4326 4328 4330 4332 4334 4336 4340 4342 4344 4346 4348 4350 -
965. 4352 4354 4356 4358 4360 4362 4364 4366 4368 4370 4372 4374 4376 TO 4394 -
966. 4396 4400 4402 4404 4406 4408 4410 4412 4414 4416 4418 4420 4422 4424 4426 -
967. 4428 4430 4432 4435 4437 4439 4441 4443 4445 4447 4449 4451 4453 4455 4457 -
968. 4459 4461 4463 4465 4467 4469 4471 4473 4475 4477 4479 4481 4483 TO 4510 -
969. 4514 4516 4518 4520 4522 4524 4526 4528 4530 4532 4534 4536 4538 4540 4542 -
970. 4544 4546 4548 4550 4552 4554 4556 4558 4560 4565 4567 4569 4571 4573 4575 -
971. 4577 4579 4581 4583 4585 4587 4589 4591 4593 4595 4597 4600 TO 4617 4619 -
972. 4621 4623 4625 4627 4629 4631 4633 4635 4637 4639 4641 4643 4645 4647 4649 -
973. 4651 4653 4655 4656 4660 4662 4664 4666 4668 4670 4672 4674 4676 4678 4680 -
974. 4682 4684 4686 4688 4690 4692 4694 4696 4698 4700 4702 4704 4706 -
975. 4708 TO 4733 4735 4737 4739 4741 4743 4745 4747 4749 4751 4753 4755
976. _RAFT_MOM 4757 4759 4761 4763 4765 4767 4769 4771 4773 4775 4777 4779 4781 -
977. 4783 4784 4787 TO 4803 4805 TO 5222 5224 TO 5240 5244 TO 5267 5269 TO 5840 -
978. 5842 TO 5865
979. _RAFT_SHEAR 4164 4168 4172 4176 4180 4184 4188 4192 4196 4200 4204 4208 4212 -
980. 4216 4220 4224 4228 4232 4256 4260 4264 4268 4272 4276 4280 4284 4288 4292 -
981. 4296 4300 4304 4308 4312 4316 4320 4324 4346 4348 4350 4352 4354 4356 4358 -
982. 4360 4362 4364 4366 4368 4370 4380 TO 4392 4404 4406 4408 4410 4412 4414 -
983. 4416 4418 4420 4422 4424 4426 4428 4441 4443 4445 4447 4449 4451 4453 4455 -
984. 4457 4459 4461 4463 4465 4467 4469 4471 4473 4475 4477 4479 4488 TO 4507 -
985. 4518 4520 4522 4524 4526 4528 4530 4532 4534 4536 4538 4540 4542 4544 4546 -
986. 4548 4550 4552 4554 4556 4569 4571 4573 4575 4577 4579 4581 4583 4585 4587 -
987. 4589 4591 4593 4603 TO 4615 4627 4629 4631 4633 4635 4637 4639 4641 4643 -
988. 4645 4647 4649 4651 4664 4666 4668 4670 4672 4674 4676 4678 4680 4682 4684 -

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 989. 4686 4688 4690 4692 4694 4696 4698 4700 4702 4711 TO 4730 4741 4743 4745 -
 990. 4747 4749 4751 4753 4755 4757 4759 4761 4763 4765 4767 4769 4771 4773 4775 -
 991. 4777 4779 4789 TO 4801 4808 TO 4820 4827 TO 4839 4843 TO 5184 5188 TO 5200 -
 992. 5207 TO 5219 5226 TO 5238 5246 TO 5265 5272 TO 5291 5298 TO 5317 -
 993. 5321 TO 5788 5792 TO 5811 5818 TO 5837 5844 TO 5863
 994. MEMBER
 995. _C1 4063 TO 4068
 996. END GROUP DEFINITION
 997. ELEMENT PROPERTY
 998. _RAFT THICKNESS 0.75
 999. DEFINE MATERIAL START
1000. ISOTROPIC CONCRETE
1001. E 2.17185E+007
1002. POISSON 0.17
1003. DENSITY 25
1004. ALPHA 1E-005
1005. DAMP 0.05
1006. TYPE CONCRETE
1007. STRENGTH FCU 27579
1008. END DEFINE MATERIAL
1009. MEMBER PROPERTY AMERICAN
1010. 8 TO 13 4063 TO 4068 PRIS YD 0.55 ZD 0.55
1011. CONSTANTS
1012. MATERIAL CONCRETE ALL
1013. SUPPORTS
1014. _RAFT PLATE MAT DIRECT Y SUBGRADE 54000
1015. **********************************************************
1016. ******************************************************************************
1017. LOAD 1 DEAD LOAD (DL)
1018. JOINT LOAD
1019. 7 FX 0 FY -57 FZ -2
1020. 8 FX 0 FY -100 FZ 0
1021. 9 FX -3 FY -126 FZ -3
1022. 10 FX 0 FY -206 FZ 3
1023. 11 FX 0 FY -85 FZ -6
1024. 12 FX 3 FY -143 FZ 7
1025. SELFWEIGHT Y -1
1026. ELEMENT LOAD
1027. _RAFT PR -13.5
1028. ******************************************************************************
1029. ******************************************************************************
1030. LOAD 2 LIVE LOAD (LL)
1031. JOINT LOAD
1032. 7 FX 0 FY -134 FZ -7
1033. 8 FX 0 FY -108 FZ 3
1034. 9 FX 0 FY -345 FZ -8
1035. 10 FX 0 FY -306 FZ 13
1036. 11 FX 0 FY -251 FZ -21
1037. 12 FX 0 FY -214 FZ 19
1038. ******************************************************************************
1039. ******************************************************************************
1040. LOAD 3 OPERATING FLUID LOAD (FL)
1041. JOINT LOAD
1042. 7 FX 0 FY 5 FZ 2
1043. 8 FX 0 FY -21 FZ 0
1044. 9 FX 6 FY -36 FZ 2
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1045. 10 FX -4 FY -168 FZ 4
1046. 11 FX 4 FY -38 FZ -4
1047. 12 FX -6 FY -197 FZ -4
1048. ******************************************************************************
1049. ******************************************************************************
1050. LOAD 4 EQP DL (EQDL)
1051. JOINT LOAD
1052. 7 FX 0 FY -26 FZ 0
1053. 8 FX 0 FY -21 FZ 0
1054. 9 FX 0 FY -52 FZ 0
1055. 10 FX 0 FY -51 FZ 0
1056. 11 FX 0 FY -22 FZ 0
1057. 12 FX 0 FY -24 FZ 0
1058. ******************************************************************************
1059. ******************************************************************************
1060. LOAD 5 EQP LL (EQLL)
1061. JOINT LOAD
1062. 7 FX 0 FY -142 FZ 0
1063. 8 FX 0 FY -128 FZ 0
1064. 9 FX 0 FY -222 FZ 0
1065. 10 FX 0 FY -223 FZ 0
1066. 11 FX 0 FY -127 FZ 0
1067. 12 FX 0 FY -121 FZ 0
1068. ******************************************************************************
1069. ******************************************************************************
1070. LOAD 6 WIND LOAD (+X) WL(+X)
1071. JOINT LOAD
1072. 7 FX 0 FY 57 FZ 0
1073. 8 FX 0 FY 38 FZ -2
1074. 9 FX 66 FY 220 FZ 0
1075. 10 FX 62 FY 242 FZ 6
1076. 11 FX 66 FY -277 FZ -3
1077. 12 FX 62 FY -279 FZ -2
1078. ******************************************************************************
1079. ******************************************************************************
1080. LOAD 7 WIND LOAD (-X) WL(-X)
1081. JOINT LOAD
1082. 7 FX 0 FY -57 FZ 0
1083. 8 FX 0 FY -38 FZ 2
1084. 9 FX -66 FY -220 FZ 0
1085. 10 FX -62 FY -242 FZ -6
1086. 11 FX -66 FY 277 FZ 3
1087. 12 FX -62 FY 279 FZ 2
1088. ******************************************************************************
1089. ******************************************************************************
1090. LOAD 8 WIND LOAD (+Z) WL(+Z)
1091. JOINT LOAD
1092. 7 FX 0 FY 169 FZ 0
1093. 8 FX 0 FY -185 FZ 64
1094. 9 FX -5 FY 222 FZ 0
1095. 10 FX 5 FY -196 FZ 86
1096. 11 FX -5 FY 170 FZ 29
1097. 12 FX 5 FY -181 FZ 29
1098. ******************************************************************************
1099. ******************************************************************************
1100. LOAD 9 WIND LOAD (-Z) WL(-Z)
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1101. JOINT LOAD
1102. 7 FX 0 FY -169 FZ 0
1103. 8 FX 0 FY 185 FZ -64
1104. 9 FX 5 FY -222 FZ 0
1105. 10 FX -5 FY 196 FZ -86
1106. 11 FX 5 FY -170 FZ -29
1107. 12 FX -5 FY 181 FZ -29
1108. ******************************************************************************
1109. ******************************************************************************
1110. LOAD 10 SEISMIC LOAD SL(+X)
1111. JOINT LOAD
1112. 7 FX 0 FY 87 FZ 0
1113. 8 FX 0 FY 66 FZ -2
1114. 9 FX 48 FY 169 FZ 0
1115. 10 FX 48 FY 211 FZ 9
1116. 11 FX 48 FY -255 FZ -4
1117. 12 FX 48 FY -276 FZ -4
1118. ******************************************************************************
1119. ******************************************************************************
1120. LOAD 11 SEISMIC LOAD SL(-X)
1121. JOINT LOAD
1122. 7 FX 0 FY -87 FZ 0
1123. 8 FX 0 FY -66 FZ 2
1124. 9 FX -48 FY -169 FZ 0
1125. 10 FX -48 FY -211 FZ -9
1126. 11 FX -48 FY 255 FZ 4
1127. 12 FX -48 FY 276 FZ 4
1128. ******************************************************************************
1129. ******************************************************************************
1130. LOAD 12 SEISMIC LOAD SL(+Z)
1131. JOINT LOAD
1132. 7 FX 0 FY 222 FZ -2
1133. 8 FX 0 FY -247 FZ 63
1134. 9 FX -9 FY 274 FZ -3
1135. 10 FX 9 FY -229 FZ 81
1136. 11 FX -9 FY 205 FZ 26
1137. 12 FX 9 FY -224 FZ 26
1138. ******************************************************************************
1139. ******************************************************************************
1140. LOAD 13 SEISMIC LOAD SL(-Z)
1141. JOINT LOAD
1142. 7 FX 0 FY -222 FZ 2
1143. 8 FX 0 FY 247 FZ -63
1144. 9 FX 9 FY -274 FZ 3
1145. 10 FX -9 FY 229 FZ -81
1146. 11 FX 9 FY -205 FZ -26
1147. 12 FX -9 FY 224 FZ -26
1148. ******************************************************************************
1149. ******************************************************************************
1150. LOAD 14 TEMPERATURE LOAD (TL)
1151. JOINT LOAD
1152. 7 FX 0 FY 0 FZ -2
1153. 8 FX 0 FY 0 FZ 0
1154. 9 FX 0 FY 7 FZ -3
1155. 10 FX 0 FY -9 FZ 6
1156. 11 FX 2 FY -7 FZ -5
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1157. 12 FX 0 FY 8 FZ 3
1158. ********************************************************************************
1159. ********************************************************************************
1160. LOAD 15 PIPE LATERAL LOAD (PL)
1161. JOINT LOAD
1162. 7 FX 0 FY 68 FZ 0
1163. 8 FX 0 FY -49 FZ 24
1164. 9 FX 24 FY 152 FZ 0
1165. 10 FX 29 FY 24 FZ 29
1166. 11 FX 24 FY 8 FZ 27
1167. 12 FX 29 FY -202 FZ 27
1168. ********************************************************************************
1169. ********************************************************************************
1170. * FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
1171. ********************************************************************************
1172. LOAD 101 1.4(DL+EQDL+FL)
1173. REPEAT LOAD
1174. 1 1.4 4 1.4 3 1.4
1175. LOAD 102 1.2(DL+EQDL+FL)+1.6(LL+EQLL)
1176. REPEAT LOAD
1177. 1 1.2 4 1.2 3 1.2 2 1.6 5 1.6
1178. LOAD 103 1.2(DL+EQDL+FL+TL)+1.6(LL+EQLL)
1179. REPEAT LOAD
1180. 1 1.2 4 1.2 3 1.2 14 1.2 2 1.6 5 1.6
1181. LOAD 104 1.4(DL+EQDL+FL+PL)
1182. REPEAT LOAD
1183. 1 1.4 4 1.4 3 1.4 15 1.4
1184. LOAD 105 1.2(DL+EQDL+FL+PL)+1.6(LL+EQLL)
1185. REPEAT LOAD
1186. 1 1.2 4 1.2 3 1.2 15 1.2 2 1.6 5 1.6
1187. LOAD 106 1.2(DL+EQDL+FL+PL+TL)+1.6(LL+EQLL)
1188. REPEAT LOAD
1189. 1 1.2 4 1.2 3 1.2 15 1.2 14 1.2 2 1.6 5 1.6
1190. LOAD 107 1.2(DL+EQDL)+0.8WL(+X)
1191. REPEAT LOAD
1192. 1 1.2 4 1.2 6 0.8
1193. LOAD 108 1.2(DL+EQDL)+0.8WL(-X)
1194. REPEAT LOAD
1195. 1 1.2 4 1.2 7 0.8
1196. LOAD 109 1.2(DL+EQDL)+0.8WL(+Z)
1197. REPEAT LOAD
1198. 1 1.2 4 1.2 8 0.8
1199. LOAD 110 1.2(DL+EQDL)+0.8WL(-Z)
1200. REPEAT LOAD
1201. 1 1.2 4 1.2 9 0.8
1202. LOAD 111 1.2(DL+EQDL)+1.6WL(+X)+0.5(LL+EQLL)
1203. REPEAT LOAD
1204. 1 1.2 4 1.2 6 1.6 2 0.5 5 0.5
1205. LOAD 112 1.2(DL+EQDL)+1.6WL(-X)+0.5(LL+EQLL)
1206. REPEAT LOAD
1207. 1 1.2 4 1.2 7 1.6 2 0.5 5 0.5
1208. LOAD 113 1.2(DL+EQDL)+1.6WL(+Z)+0.5(LL+EQLL)
1209. REPEAT LOAD
1210. 1 1.2 4 1.2 8 1.6 2 0.5 5 0.5
1211. LOAD 114 1.2(DL+EQDL)+1.6WL(-Z)+0.5(LL+EQLL)
1212. REPEAT LOAD
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1213. 1 1.2 4 1.2 9 1.6 2 0.5 5 0.5
1214. LOAD 115 1.2(DL+EQDL)+SL(+X)+0.5(LL+EQLL)+0.5FL
1215. REPEAT LOAD
1216. 1 1.2 4 1.2 10 1.0 2 0.5 5 0.5 3 0.5
1217. LOAD 116 1.2(DL+EQDL)+SL(-X)+0.5(LL+EQLL)+0.5FL
1218. REPEAT LOAD
1219. 1 1.2 4 1.2 11 1.0 2 0.5 5 0.5 3 0.5
1220. LOAD 117 1.2(DL+EQDL)+SL(+Z)+0.5(LL+EQLL)+0.5FL
1221. REPEAT LOAD
1222. 1 1.2 4 1.2 12 1.0 2 0.5 5 0.5 3 0.5
1223. LOAD 118 1.2(DL+EQDL)+SL(-Z)+0.5(LL+EQLL)+0.5FL
1224. REPEAT LOAD
1225. 1 1.2 4 1.2 13 1.0 2 0.5 5 0.5 3 0.5
1226. LOAD 119 0.9(DL+EQDL)+1.6WL(+X)
1227. REPEAT LOAD
1228. 1 0.9 4 0.9 6 1.6
1229. LOAD 120 0.9(DL+EQDL)+1.6WL(-X)
1230. REPEAT LOAD
1231. 1 0.9 4 0.9 7 1.6
1232. LOAD 121 0.9(DL+EQDL)+1.6WL(+Z)
1233. REPEAT LOAD
1234. 1 0.9 4 0.9 8 1.6
1235. LOAD 122 0.9(DL+EQDL)+1.6WL(-Z)
1236. REPEAT LOAD
1237. 1 0.9 4 0.9 9 1.6
1238. LOAD 123 0.9(DL+EQDL)+SL(+X)
1239. REPEAT LOAD
1240. 1 0.9 4 0.9 10 1.0
1241. LOAD 124 0.9(DL+EQDL)+SL(-X)
1242. REPEAT LOAD
1243. 1 0.9 4 0.9 11 1.0
1244. LOAD 125 0.9(DL+EQDL)+SL(+Z)
1245. REPEAT LOAD
1246. 1 0.9 4 0.9 12 1.0
1247. LOAD 126 0.9(DL+EQDL)+SL(-Z)
1248. REPEAT LOAD
1249. 1 0.9 4 0.9 13 1.0
1250. ************************************************
1251. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**1562**	**NUMBER OF MEMBERS**	**12**
NUMBER OF PLATES	**1470**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**1550**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 1557/ 50/ 156 DOF

TOTAL PRIMARY LOAD CASES = 41, TOTAL DEGREES OF FREEDOM = 4722
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 737 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 30.2/ 102899.5 MB

1252. CHANGE
1253. SUPPORTS
1254. _RAFT PLATE MAT DIRECT Y SUBGRADE 54000 PRINT COMPRESSION

JOINT	INFLUENCE AREA
1	0.067719
2	0.067719
3	0.067718
4	0.067718
5	0.066666
6	0.066666
1799	0.042325
1800	0.042324
1801	0.041667
1802	0.042324
1803	0.042324
1804	0.041666
1805	0.026667
1806	0.026666
1807	0.016667
1808	0.016667
1809	0.026667
1810	0.026667
1811	0.016667
1812	0.016667
1813	0.037500
1814	0.069375
1815	0.030833
1816	0.078301
1817	0.063750
1818	0.028333
1819	0.071952
1820	0.060000
1821	0.060000
1822	0.063750
1823	0.028333
1824	0.071952
1825	0.069375
1826	0.030833
1827	0.078301
1828	0.037500
1829	0.037500
1830	0.069375
1831	0.077084
1832	0.030834
1833	0.063750
1834	0.070834

1835	0.028334
1836	0.060000
1837	0.060000
1838	0.063750
1839	0.070833
1840	0.028333
1841	0.069375
1842	0.077083
1843	0.030833
1844	0.037500
1845	0.056250
1846	0.025000
1847	0.063487
1848	0.056250
1849	0.025000
1850	0.063487
1851	0.056250
1852	0.025000
1853	0.063487
1854	0.056250
1855	0.025000
1856	0.063487
1857	0.056250
1858	0.025000
1859	0.063487
1860	0.056250
1861	0.025000
1862	0.063487
1863	0.056250
1864	0.025000
1865	0.063487
1866	0.056250
1867	0.025000
1868	0.063487
1869	0.056250
1870	0.025000
1871	0.063487
1872	0.056250
1873	0.025000
1874	0.063487
1875	0.056250
1876	0.025000
1877	0.063487
1878	0.056250
1879	0.025000
1880	0.063487
1881	0.056250
1882	0.025000
1883	0.063487
1884	0.056250
1885	0.025000
1886	0.063487
1887	0.056250
1888	0.025000
1889	0.063487
1890	0.056250

1891	0.025000
1892	0.063487
1893	0.056250
1894	0.025000
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TOTAL	95.939323

1255. **
1256. ** UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
1257. **

```
1258. LOAD 201 DL+EQDL+FL
1259. REPEAT LOAD
1260. 1 1.0 4 1.0 3 1.0
1261. LOAD 202 DL+EQDL+FL+LL+EQLL+TL
1262. REPEAT LOAD
1263. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 14 1.0
1264. LOAD 203 DL+EQDL+FL+0.75(LL+EQLL)+0.75TL
1265. REPEAT LOAD
1266. 1 1.0 4 1.0 3 1.0 2 0.75 5 0.75 14 0.75
1267. LOAD 204 DL+EQDL+FL+PL
1268. REPEAT LOAD
1269. 1 1.0 4 1.0 3 1.0 15 1.0
1270. LOAD 205 DL+EQDL+FL+PL+LL+EQLL+TL
1271. REPEAT LOAD
1272. 1 1.0 4 1.0 3 1.0 15 1.0 2 1.0 5 1.0 14 1.0
1273. LOAD 206 DL+EQDL+FL+PL+0.75(LL+EQLL)+0.75TL
1274. REPEAT LOAD
1275. 1 1.0 4 1.0 3 1.0 15 1.0 2 0.75 5 0.75 14 0.75
1276. LOAD 207 DL+EQDL+FL+WL(+X)
1277. REPEAT LOAD
1278. 1 1.0 4 1.0 3 1.0 6 1.0
1279. LOAD 208 DL+EQDL+FL+WL(-X)
1280. REPEAT LOAD
1281. 1 1.0 4 1.0 3 1.0 7 1.0
1282. LOAD 209 DL+EQDL+FL+WL(+Z)
1283. REPEAT LOAD
1284. 1 1.0 4 1.0 3 1.0 8 1.0
1285. LOAD 210 DL+EQDL+FL+WL(-Z)
1286. REPEAT LOAD
1287. 1 1.0 4 1.0 3 1.0 9 1.0
1288. LOAD 211 DL+EQDL+FL+0.7SL(+X)
1289. REPEAT LOAD
1290. 1 1.0 4 1.0 3 1.0 10 0.7
1291. LOAD 212 DL+EQDL+FL+0.7SL(-X)
1292. REPEAT LOAD
1293. 1 1.0 4 1.0 3 1.0 11 0.7
1294. LOAD 213 DL+EQDL+FL+0.7SL(+Z)
1295. REPEAT LOAD
1296. 1 1.0 4 1.0 3 1.0 12 0.7
1297. LOAD 214 DL+EQDL+FL+0.7SL(-Z)
1298. REPEAT LOAD
1299. 1 1.0 4 1.0 3 1.0 13 0.7
1300. LOAD 215 DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQLL)
1301. REPEAT LOAD
1302. 1 1.0 4 1.0 3 1.0 6 0.75 2 0.75 5 0.75
1303. LOAD 216 DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQLL)
1304. REPEAT LOAD
1305. 1 1.0 4 1.0 3 1.0 7 0.75 2 0.75 5 0.75
1306. LOAD 217 DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQLL)
1307. REPEAT LOAD
1308. 1 1.0 4 1.0 3 1.0 8 0.75 2 0.75 5 0.75
1309. LOAD 218 DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQLL)
1310. REPEAT LOAD
1311. 1 1.0 4 1.0 3 1.0 9 0.75 2 0.75 5 0.75
1312. LOAD 219 DL+EQDL+FL+0.525SL(+X)+0.75(LL+EQLL)
1313. REPEAT LOAD
```

```
1314. 1 1.0 4 1.0 3 1.0 10 0.525 2 0.75 5 0.75
1315. LOAD 220 DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQLL)
1316. REPEAT LOAD
1317. 1 1.0 4 1.0 3 1.0 11 0.525 2 0.75 5 0.75
1318. LOAD 221 DL+EQDL+FL+0.525SL(+Z)+0.75(LL+EQLL)
1319. REPEAT LOAD
1320. 1 1.0 4 1.0 3 1.0 12 0.525 2 0.75 5 0.75
1321. LOAD 222 DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQLL)
1322. REPEAT LOAD
1323. 1 1.0 4 1.0 3 1.0 13 0.525 2 0.75 5 0.75
1324. LOAD 223 0.6(DL+EQDL)+WL(+X)
1325. REPEAT LOAD
1326. 1 0.6 4 0.6 6 1.0
1327. LOAD 224 0.6(DL+EQDL)+WL(-X)
1328. REPEAT LOAD
1329. 1 0.6 4 0.6 7 1.0
1330. LOAD 225 0.6(DL+EQDL)+WL(+Z)
1331. REPEAT LOAD
1332. 1 0.6 4 0.6 8 1.0
1333. LOAD 226 0.6(DL+EQDL)+WL(-Z)
1334. REPEAT LOAD
1335. 1 0.6 4 0.6 9 1.0
1336. LOAD 227 0.6(DL+EQDL)+0.7SL(+X)
1337. REPEAT LOAD
1338. 1 0.6 4 0.6 10 0.7
1339. LOAD 228 0.6(DL+EQDL)+0.7SL(-X)
1340. REPEAT LOAD
1341. 1 0.6 4 0.6 11 0.7
1342. LOAD 229 0.6(DL+EQDL)+0.7SL(+Z)
1343. REPEAT LOAD
1344. 1 0.6 4 0.6 12 0.7
1345. LOAD 230 0.6(DL+EQDL)+0.7SL(-Z)
1346. REPEAT LOAD
1347. 1 0.6 4 0.6 13 0.7
1348. ********************************************************************************
1349. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  201


**NOTE-Tension/Compression converged after  1 iterations, Case=  202


**NOTE-Tension/Compression converged after  1 iterations, Case=  203


**NOTE-Tension/Compression converged after  1 iterations, Case=  204
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 208

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

**NOTE-Tension/Compression converged after 1 iterations, Case= 211

**NOTE-Tension/Compression converged after 1 iterations, Case= 212

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

**NOTE-Tension/Compression converged after 1 iterations, Case= 214

**NOTE-Tension/Compression converged after 1 iterations, Case= 215

**NOTE-Tension/Compression converged after 1 iterations, Case= 216

**NOTE-Tension/Compression converged after 1 iterations, Case= 217

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

**NOTE-Tension/Compression converged after 1 iterations, Case= 219

**NOTE-Tension/Compression converged after 1 iterations, Case= 220

**NOTE-Tension/Compression converged after 1 iterations, Case= 221

**NOTE-Tension/Compression converged after 1 iterations, Case= 222

**NOTE-Tension/Compression converged after 1 iterations, Case= 223

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 224

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 225

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 226

**NOTE-Tension/Compression converged after 1 iterations, Case= 227

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 228

```
**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  229



**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=  230



1350. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 12,2020 TIME= 16:52:17 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

ATTACHMENT 2 STAAD INPUT & OUTPUT FILE FOR F1 STABILITY

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*                                                *
*           STAAD.Pro V8i SELECTseries5           *
*           Version  20.07.10.64                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    MAY 12, 2020                 *
*           Time=    16:56:35                     *
*                                                *
*      USER ID: LT Construction                  *
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     1. STAAD SPACE
INPUT FILE: F1 STABILITY.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 12-MAY-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. SET NL 100
     7. UNIT METER KN
     8. JOINT COORDINATES
     9. 1 0 0 0; 2 0 0 6; 3 4.9 0 0; 4 4.9 0 6; 5 11.4 0 0; 6 11.4 0 6; 7 0 2.05 0
    10. 8 0 2.05 6; 9 4.9 2.05 0; 10 4.9 2.05 6; 11 11.4 2.05 0; 12 11.4 2.05 6
    11. 1793 0 0.375 0; 1794 0 0.375 6; 1795 4.9 0.375 0; 1796 4.9 0.375 6
    12. 1797 11.4 0.375 0; 1798 11.4 0.375 6
    13. MEMBER INCIDENCES
    14. 8 1 1793; 9 2 1794; 10 3 1795; 11 4 1796; 12 5 1797; 13 6 1798; 4063 1793 7
    15. 4064 1794 8; 4065 1795 9; 4066 1796 10; 4067 1797 11; 4068 1798 12
    16. START GROUP DEFINITION
    17. ELEMENT
    18. MEMBER
    19. _C1 4063 TO 4068
    20. END GROUP DEFINITION
    21. DEFINE MATERIAL START
    22. ISOTROPIC CONCRETE
    23. E 2.17185E+007
    24. POISSON 0.17
    25. DENSITY 25
    26. ALPHA 1E-005
    27. DAMP 0.05
    28. TYPE CONCRETE
    29. STRENGTH FCU 27579
    30. END DEFINE MATERIAL
    31. MEMBER PROPERTY AMERICAN
    32. 8 TO 13 4063 TO 4068 PRIS YD 0.55 ZD 0.55
    33. CONSTANTS
    34. MATERIAL CONCRETE ALL
    35. SUPPORTS
    36. 1 TO 6 FIXED
    37. ******************************************************
    38. ********************************************************************************
```

```
 39. LOAD 1 DEAD LOAD (DL)
**WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
           MASTER/SLAVE OR IF UNCONNECTED JOINTS.
 40. JOINT LOAD
 41. 7 FX 0 FY -57 FZ -2
 42. 8 FX 0 FY -100 FZ 0
 43. 9 FX -3 FY -126 FZ -3
 44. 10 FX 0 FY -206 FZ 3
 45. 11 FX 0 FY -85 FZ -6
 46. 12 FX 3 FY -143 FZ 7
 47. SELFWEIGHT Y -1
 48. ******************************************************************************
 49. ******************************************************************************
 50. LOAD 2 LIVE LOAD (LL)
 51. JOINT LOAD
 52. 7 FX 0 FY -134 FZ -7
 53. 8 FX 0 FY -108 FZ 3
 54. 9 FX 0 FY -345 FZ -8
 55. 10 FX 0 FY -306 FZ 13
 56. 11 FX 0 FY -251 FZ -21
 57. 12 FX 0 FY -214 FZ 19
 58. ******************************************************************************
 59. ******************************************************************************
 60. LOAD 3 OPERATING FLUID LOAD (FL)
 61. JOINT LOAD
 62. 7 FX 0 FY 5 FZ 2
 63. 8 FX 0 FY -21 FZ 0
 64. 9 FX 6 FY -36 FZ 2
 65. 10 FX -4 FY -168 FZ 4
 66. 11 FX 4 FY -38 FZ -4
 67. 12 FX -6 FY -197 FZ -4
 68. ******************************************************************************
 69. ******************************************************************************
 70. LOAD 4 EQP DL (EQDL)
 71. JOINT LOAD
 72. 7 FX 0 FY -26 FZ 0
 73. 8 FX 0 FY -21 FZ 0
 74. 9 FX 0 FY -52 FZ 0
 75. 10 FX 0 FY -51 FZ 0
 76. 11 FX 0 FY -22 FZ 0
 77. 12 FX 0 FY -24 FZ 0
 78. ******************************************************************************
 79. ******************************************************************************
 80. LOAD 5 EQP LL (EQLL)
 81. JOINT LOAD
 82. 7 FX 0 FY -142 FZ 0
 83. 8 FX 0 FY -128 FZ 0
 84. 9 FX 0 FY -222 FZ 0
 85. 10 FX 0 FY -223 FZ 0
 86. 11 FX 0 FY -127 FZ 0
 87. 12 FX 0 FY -121 FZ 0
 88. ******************************************************************************
 89. ******************************************************************************
 90. LOAD 6 WIND LOAD (+X) WL(+X)
 91. JOINT LOAD
 92. 7 FX 0 FY 57 FZ 0
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 93. 8 FX 0 FY 38 FZ -2
 94. 9 FX 66 FY 220 FZ 0
 95. 10 FX 62 FY 242 FZ 6
 96. 11 FX 66 FY -277 FZ -3
 97. 12 FX 62 FY -279 FZ -2
 98. ******************************************************************************
 99. ******************************************************************************
100. LOAD 7 WIND LOAD (-X) WL(-X)
101. JOINT LOAD
102. 7 FX 0 FY -57 FZ 0
103. 8 FX 0 FY -38 FZ 2
104. 9 FX -66 FY -220 FZ 0
105. 10 FX -62 FY -242 FZ -6
106. 11 FX -66 FY 277 FZ 3
107. 12 FX -62 FY 279 FZ 2
108. ******************************************************************************
109. ******************************************************************************
110. LOAD 8 WIND LOAD (+Z) WL(+Z)
111. JOINT LOAD
112. 7 FX 0 FY 169 FZ 0
113. 8 FX 0 FY -185 FZ 64
114. 9 FX -5 FY 222 FZ 0
115. 10 FX 5 FY -196 FZ 86
116. 11 FX -5 FY 170 FZ 29
117. 12 FX 5 FY -181 FZ 29
118. ******************************************************************************
119. ******************************************************************************
120. LOAD 9 WIND LOAD (-Z) WL(-Z)
121. JOINT LOAD
122. 7 FX 0 FY -169 FZ 0
123. 8 FX 0 FY 185 FZ -64
124. 9 FX 5 FY -222 FZ 0
125. 10 FX -5 FY 196 FZ -86
126. 11 FX 5 FY -170 FZ -29
127. 12 FX -5 FY 181 FZ -29
128. ******************************************************************************
129. ******************************************************************************
130. LOAD 10 SEISMIC LOAD SL(+X)
131. JOINT LOAD
132. 7 FX 0 FY 87 FZ 0
133. 8 FX 0 FY 66 FZ -2
134. 9 FX 48 FY 169 FZ 0
135. 10 FX 48 FY 211 FZ 9
136. 11 FX 48 FY -255 FZ -4
137. 12 FX 48 FY -276 FZ -4
138. ******************************************************************************
139. ******************************************************************************
140. LOAD 11 SEISMIC LOAD SL(-X)
141. JOINT LOAD
142. 7 FX 0 FY -87 FZ 0
143. 8 FX 0 FY -66 FZ 2
144. 9 FX -48 FY -169 FZ 0
145. 10 FX -48 FY -211 FZ -9
146. 11 FX -48 FY 255 FZ 4
147. 12 FX -48 FY 276 FZ 4
148. ******************************************************************************
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149. ********************************************************************************
150. LOAD 12 SEISMIC LOAD SL(+Z)
151. JOINT LOAD
152. 7 FX 0 FY 222 FZ -2
153. 8 FX 0 FY -247 FZ 63
154. 9 FX -9 FY 274 FZ -3
155. 10 FX 9 FY -229 FZ 81
156. 11 FX -9 FY 205 FZ 26
157. 12 FX 9 FY -224 FZ 26
158. ********************************************************************************
159. ********************************************************************************
160. LOAD 13 SEISMIC LOAD SL(-Z)
161. JOINT LOAD
162. 7 FX 0 FY -222 FZ 2
163. 8 FX 0 FY 247 FZ -63
164. 9 FX 9 FY -274 FZ 3
165. 10 FX -9 FY 229 FZ -81
166. 11 FX 9 FY -205 FZ -26
167. 12 FX -9 FY 224 FZ -26
168. ********************************************************************************
169. ********************************************************************************
170. LOAD 14 TEMPERATURE LOAD (TL)
171. JOINT LOAD
172. 7 FX 0 FY 0 FZ -2
173. 8 FX 0 FY 0 FZ 0
174. 9 FX 0 FY 7 FZ -3
175. 10 FX 0 FY -9 FZ 6
176. 11 FX 2 FY -7 FZ -5
177. 12 FX 0 FY 8 FZ 3
178. ********************************************************************************
179. ********************************************************************************
180. LOAD 15 PIPE LATERAL LOAD (PL)
181. JOINT LOAD
182. 7 FX 0 FY 68 FZ 0
183. 8 FX 0 FY -49 FZ 24
184. 9 FX 24 FY 152 FZ 0
185. 10 FX 29 FY 24 FZ 29
186. 11 FX 24 FY 8 FZ 27
187. 12 FX 29 FY -202 FZ 27
188. ********************************************************************************
189. ********************************************************************************
190. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
191. ********************************************************************************
192. LOAD COMB 201 DL+EQDL+FL
193. 1 1.0 4 1.0 3 1.0
194. LOAD COMB 202 DL+EQDL+FL+LL+EQLL+TL
195. 1 1.0 4 1.0 3 1.0 2 1.0 5 1.0 14 1.0
196. LOAD COMB 203 DL+EQDL+FL+0.75(LL+EQLL)+0.75TL
197. 1 1.0 4 1.0 3 1.0 2 0.75 5 0.75 14 0.75
198. LOAD COMB 204 DL+EQDL+FL+PL
199. 1 1.0 4 1.0 3 1.0 15 1.0
200. LOAD COMB 205 DL+EQDL+FL+PL+LL+EQLL+TL
201. 1 1.0 4 1.0 3 1.0 15 1.0 2 1.0 5 1.0 14 1.0
202. LOAD COMB 206 DL+EQDL+FL+PL+0.75(LL+EQLL)+0.75TL
203. 1 1.0 4 1.0 3 1.0 15 1.0 2 0.75 5 0.75 14 0.75
204. LOAD COMB 207 DL+EQDL+FL+WL(+X)
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205. 1 1.0 4 1.0 3 1.0 6 1.0
206. LOAD COMB 208 DL+EQDL+FL+WL(-X)
207. 1 1.0 4 1.0 3 1.0 7 1.0
208. LOAD COMB 209 DL+EQDL+FL+WL(+Z)
209. 1 1.0 4 1.0 3 1.0 8 1.0
210. LOAD COMB 210 DL+EQDL+FL+WL(-Z)
211. 1 1.0 4 1.0 3 1.0 9 1.0
212. LOAD COMB 211 DL+EQDL+FL+0.7SL(+X)
213. 1 1.0 4 1.0 3 1.0 10 0.7
214. LOAD COMB 212 DL+EQDL+FL+0.7SL(-X)
215. 1 1.0 4 1.0 3 1.0 11 0.7
216. LOAD COMB 213 DL+EQDL+FL+0.7SL(+Z)
217. 1 1.0 4 1.0 3 1.0 12 0.7
218. LOAD COMB 214 DL+EQDL+FL+0.7SL(-Z)
219. 1 1.0 4 1.0 3 1.0 13 0.7
220. LOAD COMB 215 DL+EQDL+FL+0.75WL(+X)+0.75(LL+EQLL)
221. 1 1.0 4 1.0 3 1.0 6 0.75 2 0.75 5 0.75
222. LOAD COMB 216 DL+EQDL+FL+0.75WL(-X)+0.75(LL+EQLL)
223. 1 1.0 4 1.0 3 1.0 7 0.75 2 0.75 5 0.75
224. LOAD COMB 217 DL+EQDL+FL+0.75WL(+Z)+0.75(LL+EQLL)
225. 1 1.0 4 1.0 3 1.0 8 0.75 2 0.75 5 0.75
226. LOAD COMB 218 DL+EQDL+FL+0.75WL(-Z)+0.75(LL+EQLL)
227. 1 1.0 4 1.0 3 1.0 9 0.75 2 0.75 5 0.75
228. LOAD COMB 219 DL+EQDL+FL+0.525SL(+X)+0.75(LL+EQLL)
229. 1 1.0 4 1.0 3 1.0 10 0.525 2 0.75 5 0.75
230. LOAD COMB 220 DL+EQDL+FL+0.525SL(-X)+0.75(LL+EQLL)
231. 1 1.0 4 1.0 3 1.0 11 0.525 2 0.75 5 0.75
232. LOAD COMB 221 DL+EQDL+FL+0.525SL(+Z)+0.75(LL+EQLL)
233. 1 1.0 4 1.0 3 1.0 12 0.525 2 0.75 5 0.75
234. LOAD COMB 222 DL+EQDL+FL+0.525SL(-Z)+0.75(LL+EQLL)
235. 1 1.0 4 1.0 3 1.0 13 0.525 2 0.75 5 0.75
236. LOAD COMB 223 0.6(DL+EQDL)+WL(+X)
237. 1 0.6 4 0.6 6 1.0
238. LOAD COMB 224 0.6(DL+EQDL)+WL(-X)
239. 1 0.6 4 0.6 7 1.0
240. LOAD COMB 225 0.6(DL+EQDL)+WL(+Z)
241. 1 0.6 4 0.6 8 1.0
242. LOAD COMB 226 0.6(DL+EQDL)+WL(-Z)
243. 1 0.6 4 0.6 9 1.0
244. LOAD COMB 227 0.6(DL+EQDL)+0.7SL(+X)
245. 1 0.6 4 0.6 10 0.7
246. LOAD COMB 228 0.6(DL+EQDL)+0.7SL(-X)
247. 1 0.6 4 0.6 11 0.7
248. LOAD COMB 229 0.6(DL+EQDL)+0.7SL(+Z)
249. 1 0.6 4 0.6 12 0.7
250. LOAD COMB 230 0.6(DL+EQDL)+0.7SL(-Z)
251. 1 0.6 4 0.6 13 0.7
252. **********************************************************************
253. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**18**	**NUMBER OF MEMBERS**	**12**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**6**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 12/ 1/ 12 DOF
TOTAL PRIMARY LOAD CASES = 15, TOTAL DEGREES OF FREEDOM = 72
TOTAL LOAD COMBINATION CASES = 30 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 102853.6 MB

254. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 12,2020 TIME= 16:56:40 ****

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*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
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